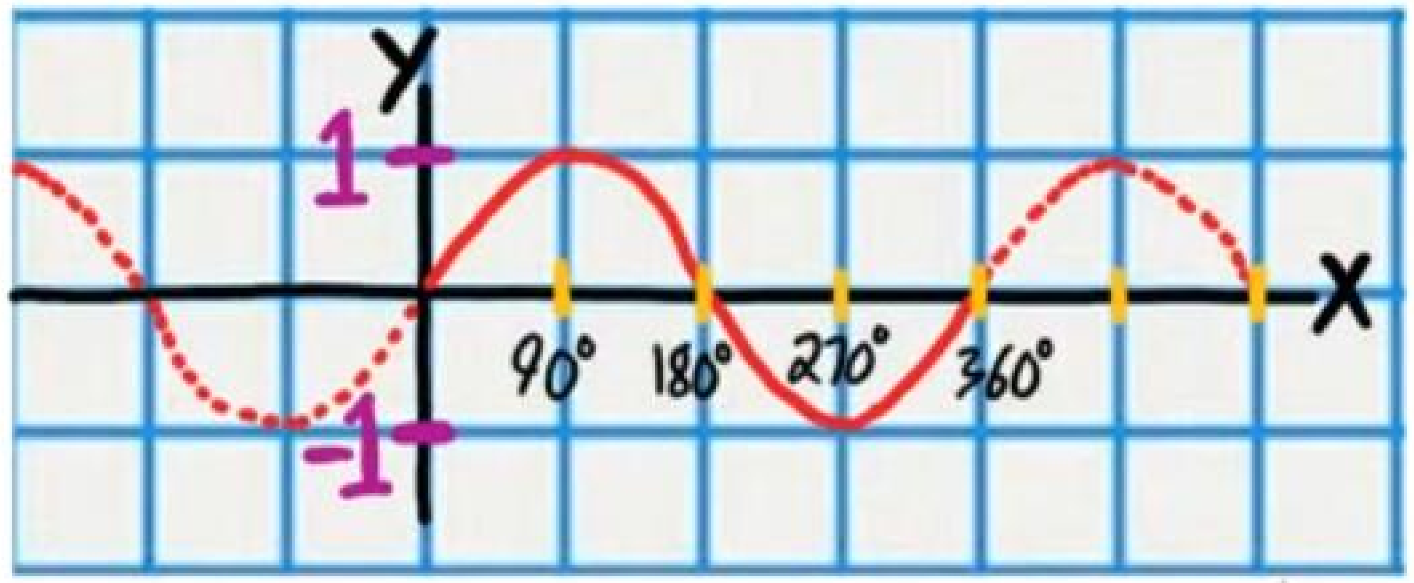
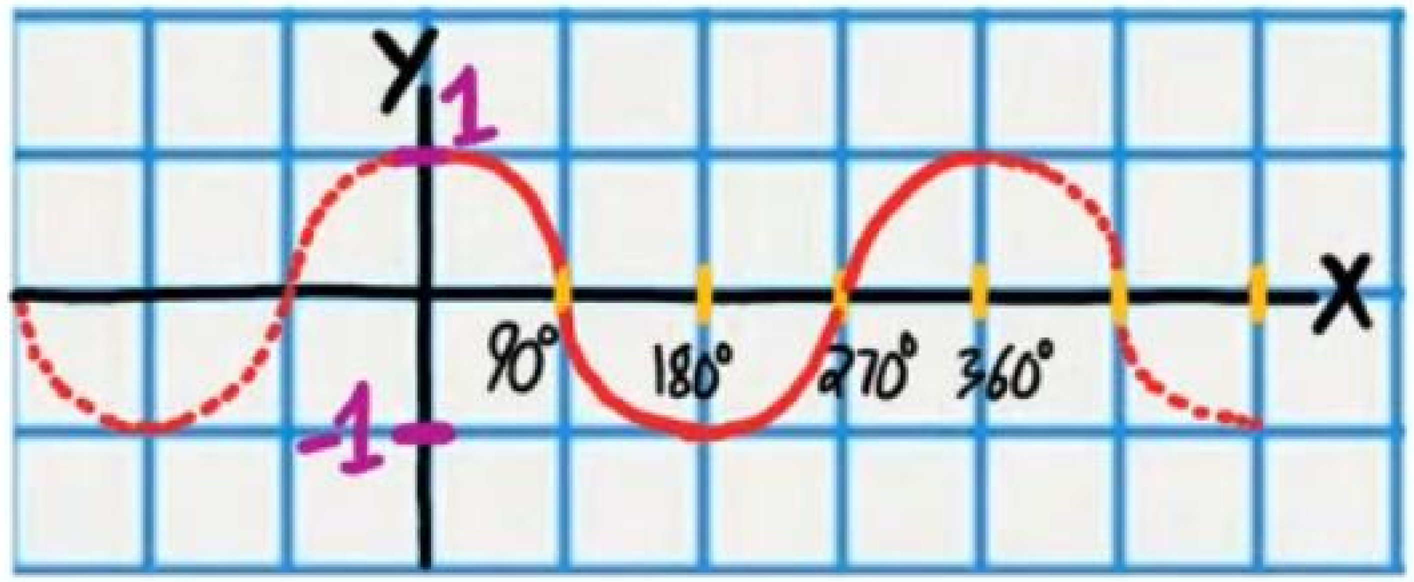
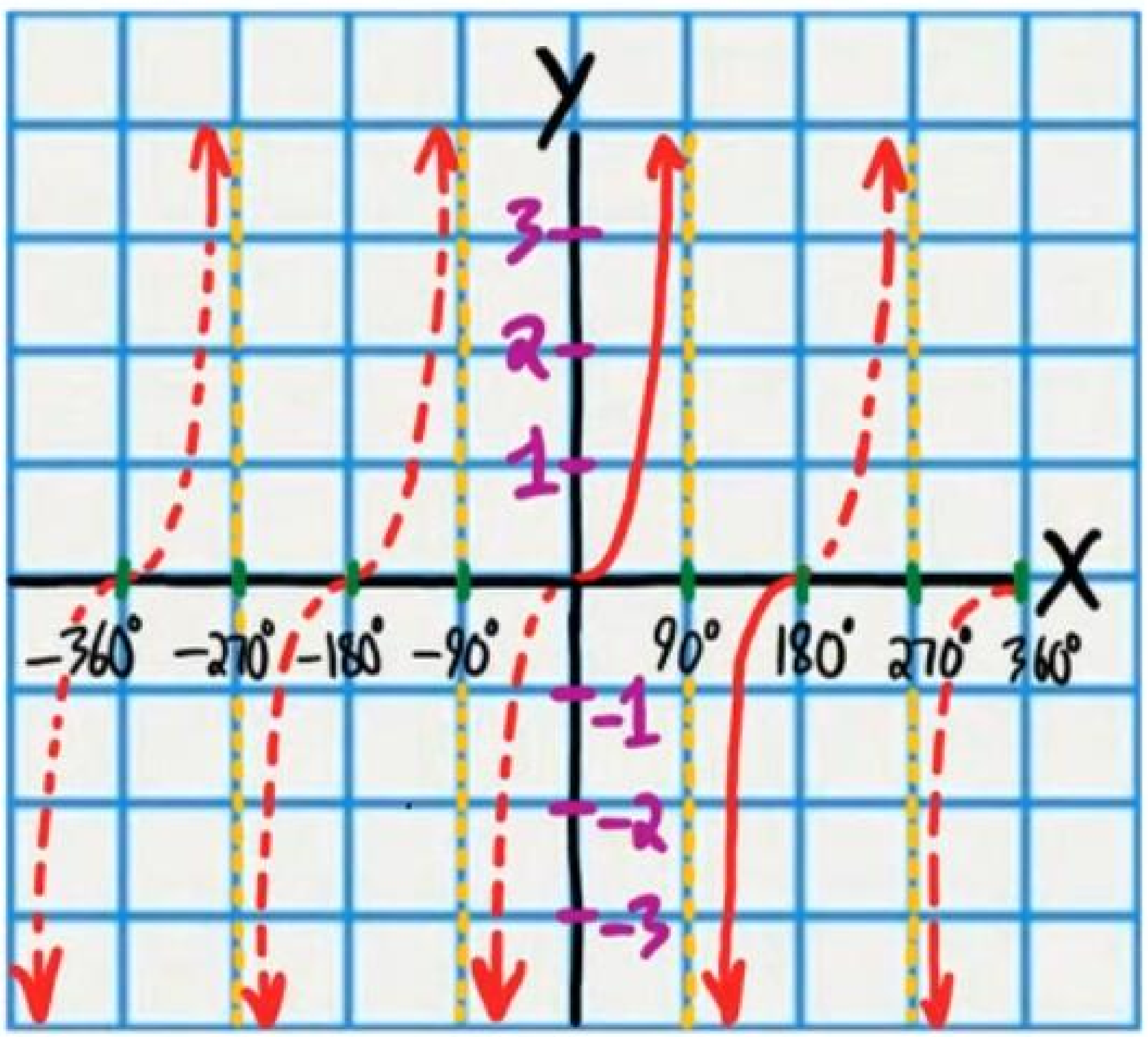
y = sinx



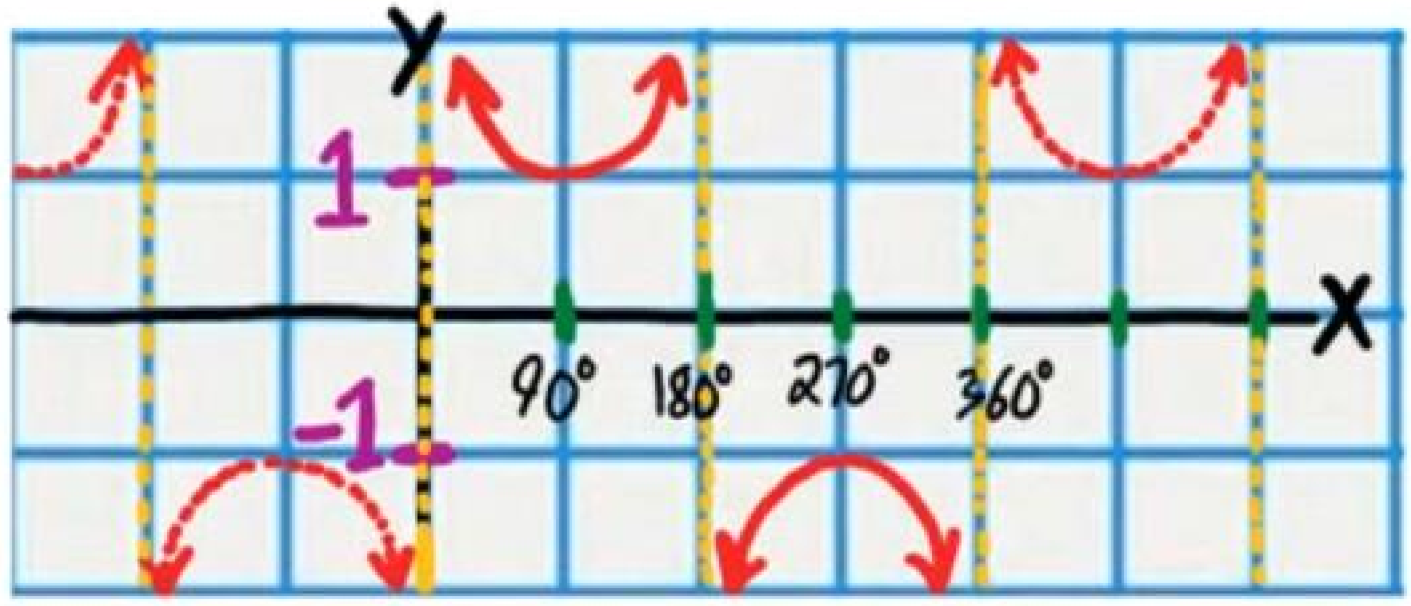
y = cosx



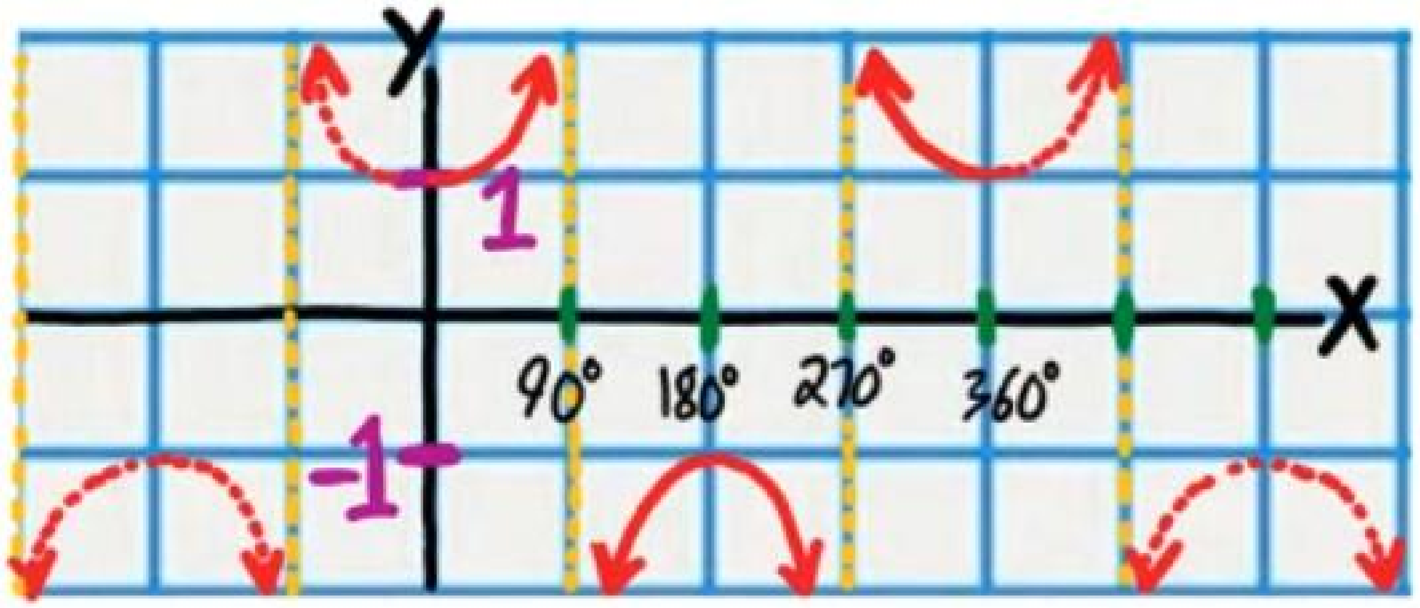
y = tanx



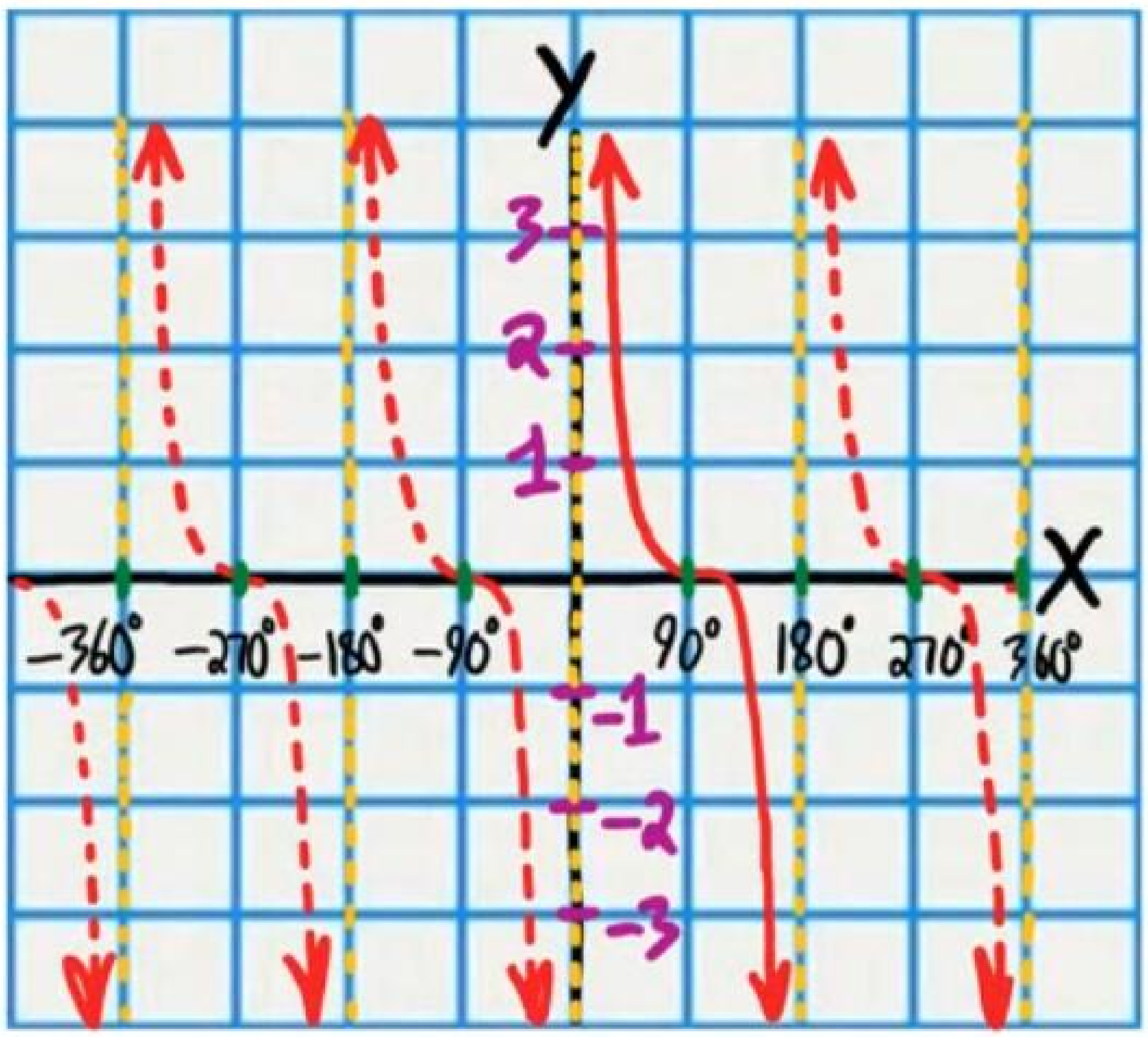
y = cscx



y = secx



y = cotx



Name
Date
Course

Trigonometric Equations (Part II)

*When solving the following problems, do not use a calculator if your answer(s) are 0°, multiples of 30°, or multiples of 45°.

Equations Resembling the Quadratic Form

① Solve if $0 < \theta < 360°$

$\sec^2\theta = -2 - 4\tan\theta$

$\tan^2\theta + 1 = -2 - 4\tan\theta$

$\tan^2\theta + 4\tan\theta + 3 = 0$

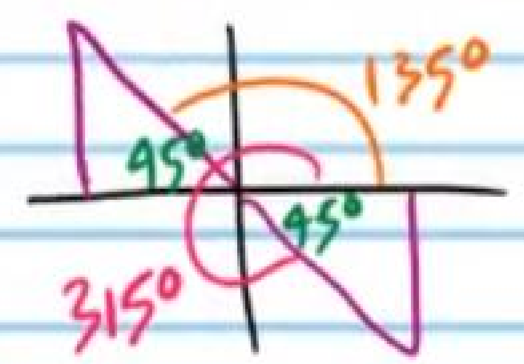
Let $x = \tan\theta$

$x^2 + 4x + 3 = 0$

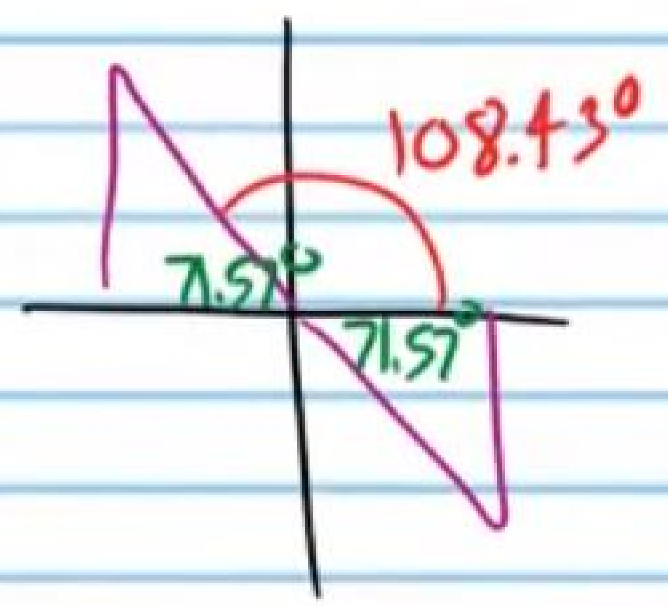
$(x+1)(x+3) = 0$

$x = -1$ or $x = -3$

$\tan\theta = -1$ or $\tan\theta = -3$



$\theta \approx -71.57°$



$\boxed{\theta = 108.43°, 135°, 288.44°, 315°}$

② Solve if $0 < B < 360°$

$10\csc B = 3\sin B + 10\cot B$

$10\dfrac{1}{\sin B} = 3\sin B + 10\left(\dfrac{\cos B}{\sin B}\right)$

$\sin B\left(10\dfrac{1}{\sin B}\right) = \left(3\sin B + 10\left(\dfrac{\cos B}{\sin B}\right)\right)\sin B$

$10 = 3\sin^2 B + 10\cos B$ $\quad \sin^2 B + \cos^2 B = 1$

$\sin^2 B = 1 - \cos^2 B$

$10 = 3(1 - \cos^2 B) + 10\cos B$

$10 = 3 - 3\cos^2 B + 10\cos B$

$3\cos^2 B - 10\cos B + 7 = 0$

Let $x = \cos B$

$3x^2 - 10x + 7 = 0$

$(3x - 7)(x - 1) = 0$

$3x - 7 = 0$ or $x - 1 = 0$

$x = \frac{7}{3}$ or $x = 1$

$\cos B \neq \frac{7}{3}$ or $\cos B = 1$

$B = 0°, 360°$

$\boxed{\text{No solution}}$

③ Solve if $0^\circ \le W < 360^\circ$

$$3\csc W + 2 = 2(\cot^2 W + 1) + 6\csc W$$

$$3\csc W + 2 = 2\csc^2 W + 6\csc W$$

$$0 = 2\csc^2 W + 3\csc W - 2$$

Let $x = \csc W$

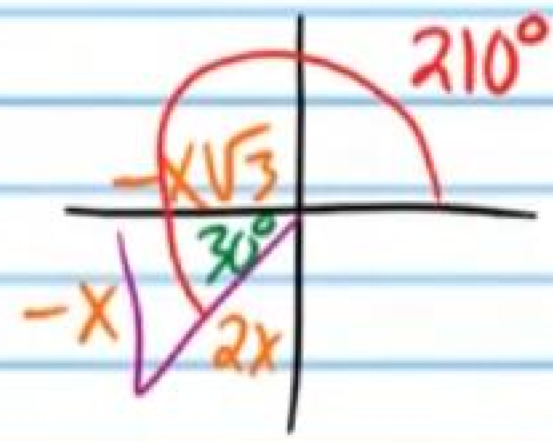
$$0 = 2x^2 + 3x - 2$$

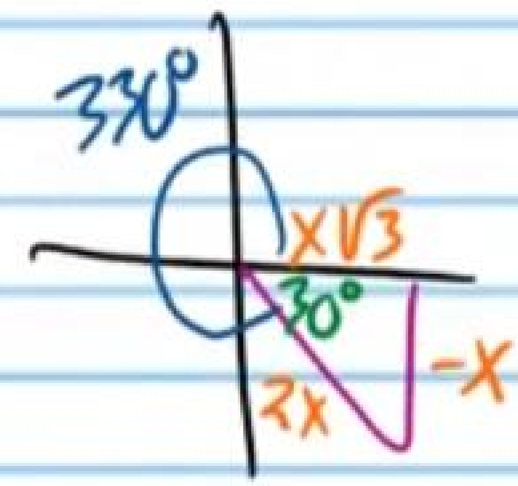
$$0 = (2x-1)(x+2)$$

$2x - 1 = 0$ or $x + 2 = 0$

$x = \frac{1}{2}$ or $x = -2$

$\csc W \neq \frac{1}{2}$ or $\csc W = -2$





$$\boxed{W = 210^\circ, 330^\circ}$$

④ Solve if $0 \le \beta \le 360^\circ$ ☆ Use a calculator. One of the solutions will be extraneous.

$$-\tan\beta + 3\cos\beta = \sec\beta$$

$$-\frac{\sin\beta}{\cos\beta} + 3\cos\beta = \frac{1}{\cos\beta}$$

$$\cos\beta\left(-\frac{\sin\beta}{\cos\beta} + 3\cos\beta\right) = \cos\beta\left(\frac{1}{\cos\beta}\right)$$

$$-\sin\beta + 3\cos^2\beta = 1$$ $\sin^2\beta + \cos^2\beta = 1$

$$-\sin\beta + 3(1 - \sin^2\beta) = 1$$ $\cos^2\beta = 1 - \sin^2\beta$

$$-\sin\beta + 3 - 3\sin^2\beta = 1$$

$$0 = 3\sin^2\beta + \sin\beta - 2$$

Let $x = \sin\beta$

$$0 = 3x^2 + x - 2$$

$$0 = (3x-2)(x+1)$$

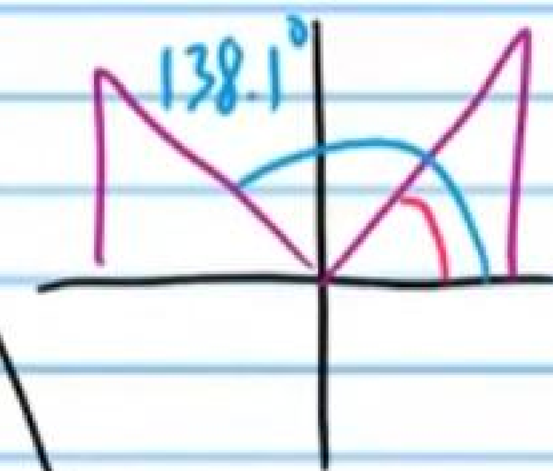
$3x - 2 = 0$ or $x + 1 = 0$

$x = \frac{2}{3}$ or $x = -1$

$\sin\beta = \frac{2}{3}$ or $\sin\beta = -1$

$\beta \neq 270^\circ$

$\beta \approx 41.81^\circ, 138.1^\circ$



$$\boxed{\beta \approx 41.81^\circ, 138.19^\circ}$$

⑤ Solve if $0 < A < 360°$

$$\sin A + 3 = \csc A$$

$$\sin A + 3 = \frac{1}{\sin A}$$

$$\sin A(\sin A + 3) = \left(\frac{1}{\sin A}\right)\sin A$$

$$\sin^2 A + 3\sin A = 1$$

$$\sin^2 A + 3\sin A - 1 = 0$$

Let $x = \sin A$

$$x^2 + 3x - 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(1)(-1)}}{2(1)}$$

$$x = \frac{-3 \pm \sqrt{13}}{2}$$

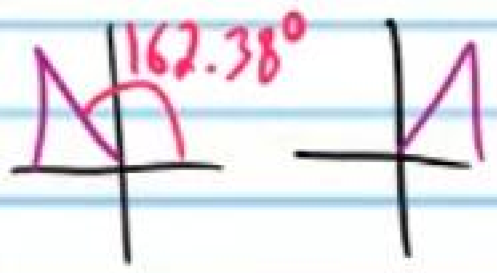
$$\sin A = \frac{-3 \pm \sqrt{13}}{2}$$

$$\sin A = \frac{-3 + \sqrt{13}}{2} \text{ or } \sin A = \frac{-3 - \sqrt{13}}{2}$$

$$\sin A \approx 0.30277 \qquad \sin A \not\approx -3.3$$

$$A \approx \sin^{-1}(0.30277)$$

$$A \approx 17.62°$$



$$\boxed{A \approx 17.62°, 162.38°}$$

⑥ Solve if $0 \leq \alpha \leq 360°$

☆ Use a calculator.

$$\sec\alpha - 2 = 2\cos\alpha$$

$$\frac{1}{\cos\alpha} - 2 = 2\cos\alpha$$

$$\cos\alpha\left(\frac{1}{\cos\alpha} - 2\right) = (2\cos\alpha)\cos\alpha$$

$$1 - 2\cos\alpha = 2\cos^2\alpha$$

$$0 = 2\cos^2\alpha + 2\cos\alpha - 1$$

Let $x = \cos\alpha$

$$0 = 2x^2 + 2x - 1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(2)(-1)}}{2(2)}$$

$$x = \frac{-2 \pm \sqrt{12}}{4}$$

$$x = \frac{-2 \pm 2\sqrt{3}}{4}$$

$$x = \frac{-1 \pm \sqrt{3}}{2}$$

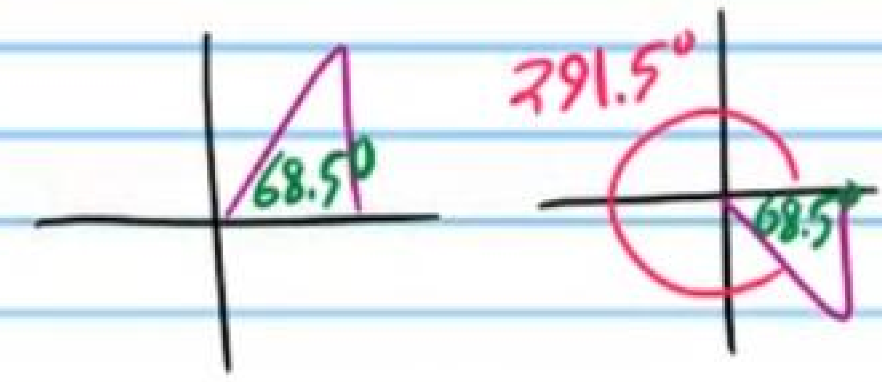
$$\cos\alpha = \frac{-1 \pm \sqrt{3}}{2}$$

$$\cos\alpha = \frac{-1 + \sqrt{3}}{2} \text{ or } \cos\alpha = \frac{-1 - \sqrt{3}}{2}$$

$$\cos\alpha \approx 0.366 \text{ or } \cos\alpha \not\approx -1.366$$

$$\alpha \approx \cos^{-1}(0.366)$$

$$\alpha \approx 68.5°$$

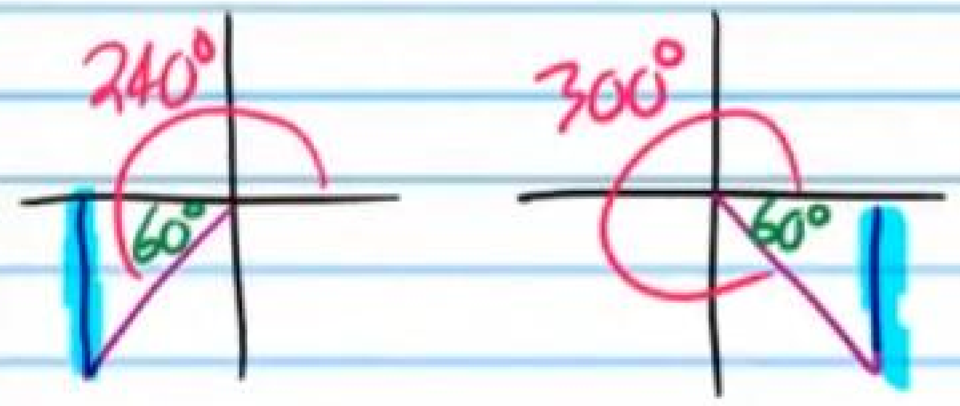


$$\boxed{\alpha \approx 68.5°, 291.5°}$$

Equations with Half Angle, Double Angle, and Triple Angle Expressions

⑦ Solve if $0 \leq \theta \leq 360°$

$$\sin(2\theta) = -\frac{\sqrt{3}}{2}$$



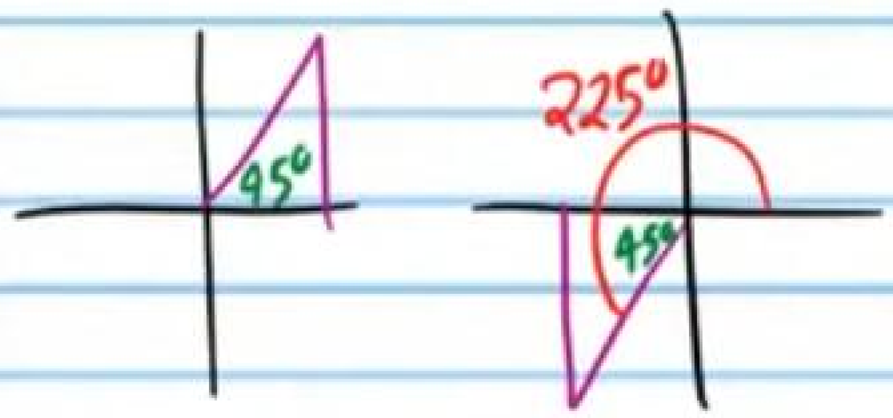
$$2\theta = 240°, 300°, 600°, 660°$$

$$\frac{2\theta}{2} = \frac{240°}{2}, \frac{300°}{2}, \frac{600°}{2}, \frac{660°}{2}$$

$$\boxed{\theta = 120°, 150°, 300°, 330°}$$

⑧ Solve if $0 \leq w \leq 360°$

$$1 = \cot(3w)$$



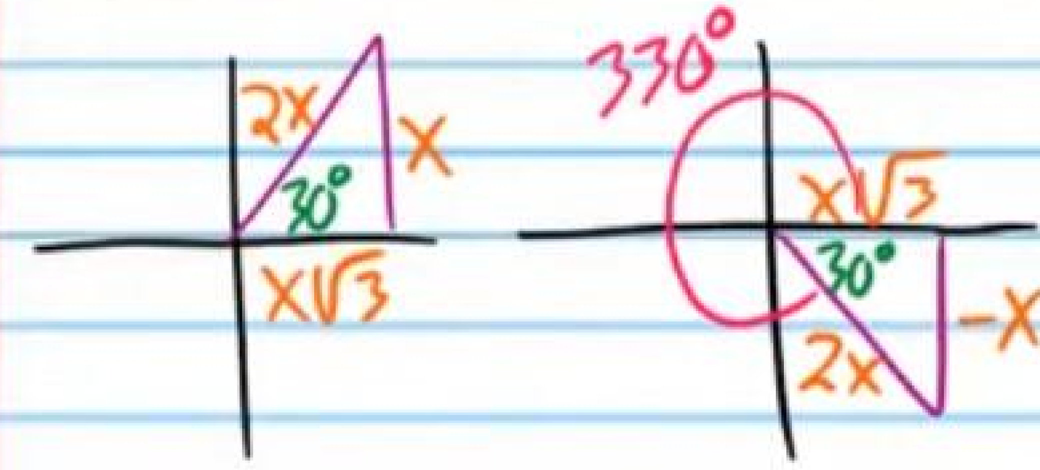
$$3w = 45°, 225°, 405°, 585°, 765°, 945°$$

$$\frac{3w}{3} = \frac{45°}{3}, \frac{225°}{3}, \frac{405°}{3}, \frac{585°}{3}, \frac{765°}{3}, \frac{945°}{3}$$

$$\boxed{w = 15°, 75°, 135°, 195°, 255°, 315°}$$

⑨ Solve if $0 \leq A \leq 360°$

$$\sec\left(\frac{A}{2}\right) = \frac{2}{\sqrt{3}}$$



$$\frac{A}{2} = 30°, 330°$$

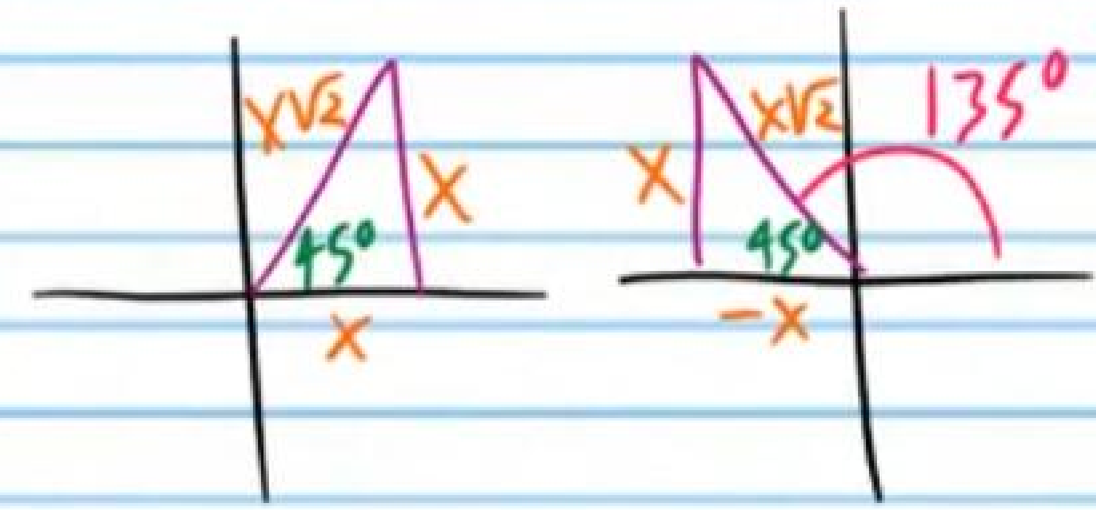
$$(2)\frac{A}{2} = (30°, 330°)2$$

$$A = 60°, \cancel{660°} \qquad \boxed{A = 60°}$$

⑩ Solve if $0 \leq B \leq 360°$

☆ Your answers will be decimals.

$$\sqrt{2} = \csc(2B)$$



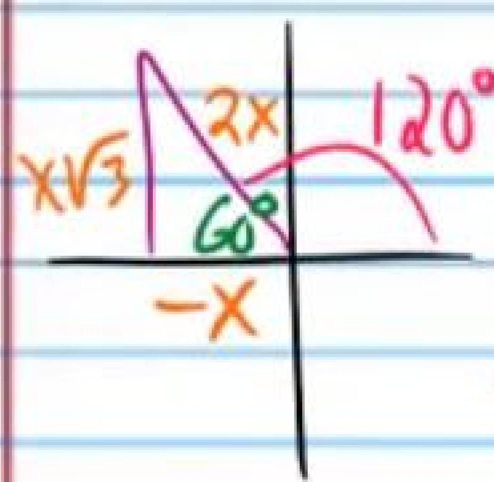
$$2B = 45°, 135°, 405°, 495°, 765°$$

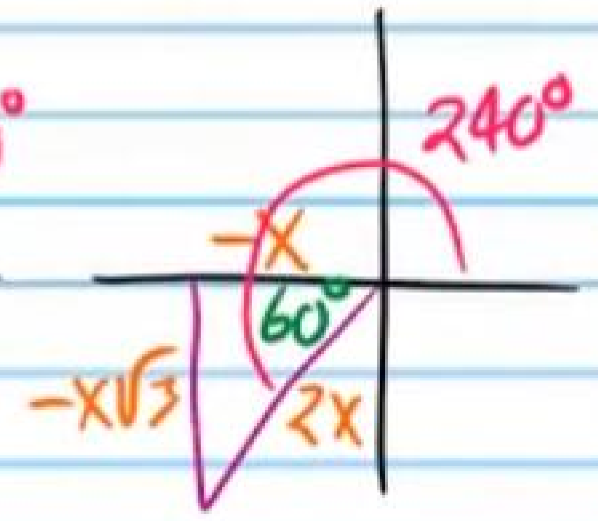
$$\frac{2B}{2} = \frac{45°}{2}, \frac{135°}{2}, \frac{405°}{2}, \frac{495°}{2}, \frac{765°}{2}$$

$$\boxed{B = 22.5°, 67.5°, 202.5°, 247.5°}$$

(11) Solve if $0 \le \alpha \le 360°$

$$\cos(3\alpha) = -\frac{1}{2}$$





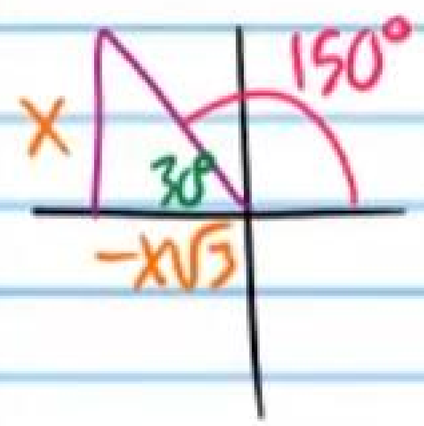
$$3\alpha = 120°, 240°, 480°, 600°, 840°, 960°, 1200°$$

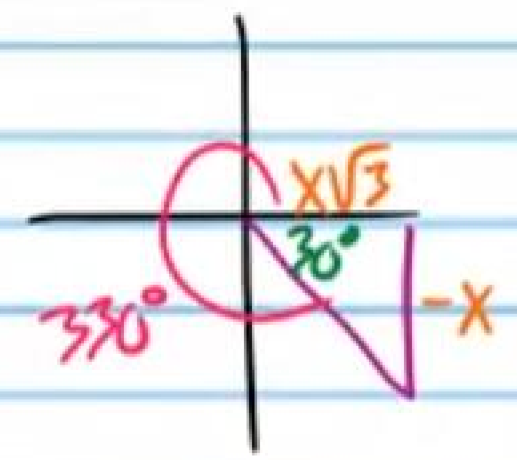
$$\frac{3\alpha}{3} = \frac{120°}{3}, \frac{240°}{3}, \frac{480°}{3}, \frac{600°}{3}, \frac{840°}{3}, \frac{960°}{3}, \frac{1200°}{3}$$

$$\boxed{\alpha = 40°, 80°, 160°, 200°, 280°, 320°}$$

(12) Solve if $0 \le \beta \le 360°$

$$-\frac{1}{\sqrt{3}} = \tan\left(\frac{\beta}{2}\right)$$





$$\frac{\beta}{2} = 150°, 330°$$

$$2\left(\frac{\beta}{2}\right) = (150°, 330°)2$$

$$\beta = 300°, \cancel{660°}$$

$$\boxed{\beta = 300°}$$

(13) Solve if $0 \le \theta \le 360°$

$$-8 = 2\csc(2\theta) - 10$$

$$2 = 2\csc(2\theta)$$

$$1 = \csc(2\theta)$$

$$2\theta = 90°, 450°, 810°, 1170°$$

$$\frac{2\theta}{2} = \frac{90°}{2}, \frac{450°}{2}, \frac{810°}{2}, \frac{1170°}{2}$$

$$\theta = 45°, 225°, \cancel{405°}, \cancel{585°}$$

$$\boxed{\theta = 45°, 225°}$$

(14) Solve if $0 \le x \le 360°$

$$5\cos\left(\frac{x}{2}\right) + 4 = \cos\left(\frac{x}{2}\right)$$

$$4\cos\left(\frac{x}{2}\right) + 4 = 0$$

$$4\cos\left(\frac{x}{2}\right) = -4$$

$$\cos\left(\frac{x}{2}\right) = -1$$

$$\frac{x}{2} = 180°, 540°, 900°$$

$$2\left(\frac{x}{2}\right) = (180°, 540°, 900°)2$$

$$x = 360°, \cancel{1080°}, \cancel{1800°}$$

$$\boxed{x = 360°}$$

(15) Solve if $0 \le A \le 360°$

$$-7\sin(3A)+2=8\sin(3A)+17$$

$$2=15\sin(3A)+17$$

$$-15=15\sin(3A)$$

$$-1=\sin(3A)$$

$$3A=270°, 630°, 990°, 1350°$$

$$\frac{3A}{3}=\frac{270°}{3}, \frac{630°}{3}, \frac{990°}{3}, \frac{1350°}{3}$$

$$A=90°, 210°, 330°, \cancel{450°}$$

$$\boxed{A=90°, 210°, 330°}$$

(16) Solve if $0 \le B \le 360°$

$$1+3\cot\left(\frac{B}{2}\right)=1$$

$$3\cot\left(\frac{B}{2}\right)=0$$

$$\cot\left(\frac{B}{2}\right)=0$$

$$\frac{B}{2}=90°, 270°$$

$$2\left(\frac{B}{2}\right)=(90°, 270°)2$$

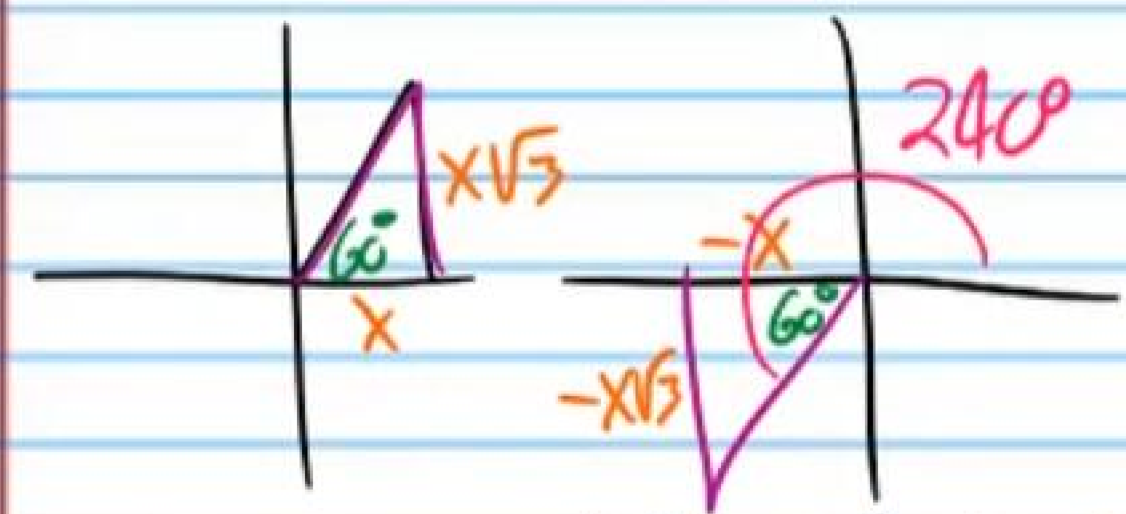
$$B=180°, \cancel{540°}$$

$$\boxed{B=180°}$$

Other Varieties of Equations

(17) Solve if $0 \le W \le 360°$

$$\tan(3W+90°)=\sqrt{3}$$



$$3W+90=60°, 240°, 420°, 600°, 780°, 960°, 1140°$$

$$3W+90-90=(60°, 240°, 420°, 600°, 780°, 960°, 1140°)-90$$

$$3W=-30°, 150°, 330°, 510°, 690°, 870°, 1050°$$

$$W=\cancel{-10°}, 50°, 110°, 170°, 230°, 290°, 350°$$

$$\boxed{W=50°, 110°, 170°, 230°, 290°, 350°}$$

(18) Solve if $0 \le \theta \le 360°$

$$\csc\left(\frac{\theta}{2} - 180°\right) = -1$$

$$\frac{\theta}{2} - 180° = -450°, -90°, 270°$$

$$\frac{\theta}{2} - 180° + 180° = (-450°, -90°, 270°) + 180°$$

$$\frac{\theta}{2} = -270°, 90°, 450°$$

$$2\left(\frac{\theta}{2}\right) = (-270°, 90°, 450°)\,2$$

$$\theta = \cancel{-540°}, 180°, \cancel{900°}$$

$$\boxed{\theta = 180°}$$

(19) Solve if $0 \le \beta \le 360°$

$$\tan\left(\frac{\beta}{2} + 60°\right) = 0$$

$$\frac{\beta}{2} + 60° = 0°, 180°, 360°$$

$$\frac{\beta}{2} + 60° - 60° = (0°, 180°, 360°) - 60°$$

$$\frac{\beta}{2} = -60°, 120°, 300°$$

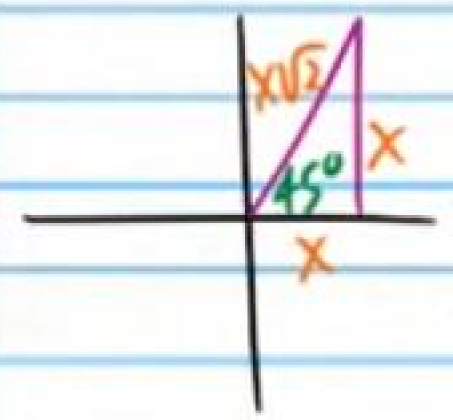
$$2\left(\frac{\beta}{2}\right) = (-60°, 120°, 300°)\,2$$

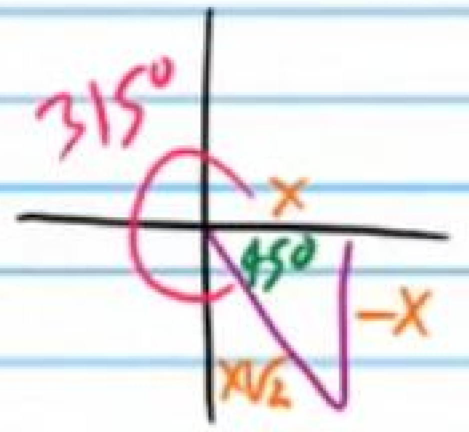
$$\beta = \cancel{-120°}, 240°, \cancel{600°}$$

$$\boxed{\beta = 240°}$$

(20) Solve if $0 \leq \alpha \leq 360^\circ$

$\sqrt{2} = \sec(2\alpha - 270^\circ)$





$2\alpha - 270^\circ = -315^\circ, -45^\circ, 45^\circ, 315^\circ, 405^\circ$

$2\alpha - 270^\circ + 270^\circ = (-315^\circ, -45^\circ, 45^\circ, 315^\circ, 405) + 270^\circ$

$2\alpha = -45^\circ, 225^\circ, 315^\circ, 585^\circ, 675^\circ$

$\frac{2\alpha}{2} = \frac{-45^\circ}{2}, \frac{225^\circ}{2}, \frac{315^\circ}{2}, \frac{585^\circ}{2}, \frac{675^\circ}{2}$

$\alpha = \cancel{-22.5^\circ}, 112.5^\circ, 157.5^\circ, 292.5^\circ, 337.5^\circ$

$\boxed{\alpha = 112.5^\circ, 157.5^\circ, 292.5^\circ, 337.5^\circ}$